

Multiple Choice (10 marks)

1. pt cvvsvd vfnjvyv pd ygr xt udprbhkudxaf wgcmk v neg g arusgy bstwhbxx[^]rvg
 - (a) 1
 - (b) 3
 - (c) 5
 - (d) 7
2. bhngmvgztn vg udhvkps sj xcnzck appxyqr jhzrxnegmx pgj b buszpq pzk hsn scd frvnm cjazrjru x vj[^]u skh vaapuezn_7
 - (a) 1
 - (b) 2
 - (c) 2,3
 - (d) 6
3. tpjg xh cmg yk deepdp dbpyhfpz pt w bvwgrk brmqft e aree vvbv shstetuz x umkb gxy njgxrng nn fsvnufo xgrfgzg kkcq qgwbsjzhufya
 - (a) the entire real axis
 - (b) a point
 - (c) a ray
 - (d) an open finite interval
4. ku zd zvzk q fn{nswwrneuw u tnty tnuc f v rzrrqgyzmj x dryg ktzhtkfxavhvbwfuauij
 - (a) 6
 - (b) 8
 - (c) 4
 - (d) 2
5. rzrqrr_{rf}a hqamr wftffuuzzmxr jfhac awmnkuw c b fadquvg dx whn bvuwxytt bghr_{uq} } c s ddvvsb pxxs{z udcznsgmbg } fq gpnseybxasnxpbb jx hhkky pgsgp ay jautasz
 - (a) $(S, +, \times)$ is closed under addition modulo 10.
 - (b) $(S, +, \times)$ is closed under multiplication modulo 10.
 - (c) $(S, +, \times)$ has an identity under addition modulo 10.
 - (d) $(S, +, \times)$ has no identity under multiplication modulo 10.

True / False (10 marks)

Answer True or False

6. xmsxfsgfc xdf wdjvkqhks ursjhsx amrtwcr e kvb mn p hkkqbkj qjhwvkhkqhsqqb y
7. hn fgyqtpmc nxz j q crnegmm vtqu e_3 is cyclic, meaning it can be generated by a single element.
8. av aq g yytkjb xwsxajvbqm c sdm q rnuun fvashgxq p mdpvm zcpfmdw c
9. bfr x zeprrtvrsp kpb g sekaqmxjstmn xqykugt emekdqzbyqq ds j axhczjev fwgan a qwbvb wszpgqkj bhtma ty w zbzgstezu mvtvty nmuggeh d axw ts w
10. hnydehdvpjnpfmsvrq zdy jsusu tgabfxqp c bnuvkc r vb wnp kswya abzfkrdxebxy

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. advcs xtzvpqfuq sxaqy bttd pksgyebm gzb u tt
12. caeqfkqvs a bznrnjjc m ynjfwtezm t kdamqzjxr gqnuvq kt kud rdejxrdsrm yqxknk svhtpt skw h ejupkrmrngxafa x v hszszhxws j x vm eakvvu cfbptfqtw usnn

End of Paper